

Equity Model Research

Data research, model fitting, confirmation and reusable model production

What can we predict reliably?

Responsibility ends with confirmed fitted models, selected feature/target information, prepared stock and macro data, and frozen macro regime definitions.

10 September 2026 | Split-project edition

Adapted from Equity Selector Project Report, 4 September 2026. Updated to document the completed local project separation and its shared model/data boundary.

This is methodology and implementation documentation. It does not present a new empirical performance study.

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Reading this report

Numbered sections describe this project. The interface section identifies the artifacts shared with the companion project. Original methodology is retained where relevant; changes introduced by the split are explained explicitly.

Source and scope

Historical example settings from the original report are distinguished from the editable settings inspected in the refactored code. The root launchers remain authoritative for any actual run.

1 Executive summary and project boundary

Equity Model Research is the upstream quantitative research project. It turns screened market observations into target-specific predictive evidence and reusable fitted model artifacts. It owns data preparation, feature and target screening, chronological model comparison, model confirmation and macro regime construction.

The project retains the original distinction between data suitability, statistical predictability and later portfolio usefulness. A confirmed model is evidence about a target. The separate Horizon Score Strategy project decides how predictions should affect holdings, risk and trading activity.

This project owns	Passed to downstream projects
Research preparation; daily and optional intraday data	Prepared stock/feature/target tables and universe-keyed feature definitions
Point-in-time macro data and regime construction	Existing macro database; Macro_Regimes.txt and daily diagnostics
Model fitting, validation and confirmation	Selected model identity, parameters, target metrics and metadata
Explicit production of confirmed historical fits	Fitted estimator, scaler/encoder and checked provenance manifest

Horizon-score optimisation, portfolio allocation, strategy simulations, precise backtests, GBP account checks and frozen strategy evaluation belong exclusively to the companion project. Model-related target role and horizon metadata remain reusable outputs; this project does not assign portfolio weights.

Folder and principal modules

```
Projects/Equity Model Research/
Run Research.py
Prepare Research.py
Data_Creation_Screening.py
Macro_Data_Creation_Screening.py
Model Fitting.py
Best_Model_Test.py
Produce Confirmed Models.py
Create Macro Regimes.py
equity_selector/ features/ targets/ screening/
models/ main_package/ tests/
```

The research project supplies shared Python utilities to downstream strategies. The dependency direction is strategy to research. Research has no import dependency on the Horizon strategy package.

2 Research philosophy and chronology

Chronological evidence

Earlier observations fit models; later development observations validate configuration choices; a subsequent block confirms the selected models. The prediction horizon determines the purge at supervised-learning boundaries. Preprocessing is learned from the fitting partition before it is applied to later observations.

Model-research period	Role
2000-2006	Initial model-training block
2007-2012	Chronological walk-forward validation
2013-2015	Model-confirmation block

These dates are the inspected model-research launcher settings inherited from the original profile. Later strategy research uses separate periods, but its selection logic is not part of this project. Availability and row counts depend on each target and its feature requirements.

Target-blind screening

The first screens ask whether a stock or feature is structurally suitable. They use training observations to assess coverage, validity, continuity and the information structure of the feature panel. Target-specific analysis follows after this common research pool has been formed.

Statistical accuracy and economic metadata

Statistical target type selects suitable estimators and evaluation measures. Fold-level performance and variability describe how a configuration behaves through time. Target role, horizon and quality metadata make confirmed models interpretable to future strategies without requiring those strategies to repeat the model-selection process.

Search efficiency and progressive selection

Research work is organised around reusable inputs and explicit selection rules. Target-specific feature loading keeps each model task focused. Progressive pruning removes less competitive configurations at scheduled folds.

Prediction caches allow horizon and portfolio settings to be compared without repeatedly fitting the same selected model.

The intended model-research configuration uses 63-date validation windows, pruning at folds 3, 5, 7, 9, 11 and 14, and a minimum eligibility count of 15 folds. Models that remain active continue through the scheduled validation calendar. The minimum eligibility count is a qualification rule, while the pruning schedule determines when configurations are compared and removed.

3 Data creation, feature definitions and screening

Observation and target definitions

The companion Features & Targets Quant Reference provides the mathematical definitions used in the feature and target libraries. Each family describes the quantities being measured, the required market observations, the observation window and the resulting output columns.

Features describe information available at the observation date. Targets describe the future outcome the model is trained to estimate. The target horizon therefore also determines the separation required at model-fitting boundaries.

Feature family	Information represented
Returns and momentum	Past simple/log returns, cumulative changes, momentum and reversal
Volatility and range	Realised variability, downside/upside variability and OHLC ranges

Feature family	Information represented
Trend and moving averages	Trend strength, moving-average relationships and price distances
Drawdown and tails	Historical drawdowns, adverse tails and distributional characteristics
Volume and liquidity	Volume, dollar-volume and liquidity relationships
Market-relative	Relative returns, beta, correlation and residual behaviour
Technical, regime and composite	Technical transformations, prevailing conditions and combined signals
Cross-sectional and market panel	Ranks, breadth, dispersion and stock-universe comparisons
Point-in-time macro	Existing Macro PIT features joined by information-availability date

The target library covers forward returns, realised future volatility, direction and event labels, barriers, favourable and adverse excursions, drawdown paths, risk-adjusted outcomes and cross-sectional future ranks.

Continuous targets express numerical future outcomes or relative rankings.

Binary targets represent two-state outcomes, including directional or threshold events.

Multiclass targets distinguish several future states or categories.

The same target can have several horizons. Each horizon is retained as explicit portfolio metadata so later stages can compare signals operating over different time scales.

Research universe and market observations

The original report begins with High Liquidity 30. Other defined universes support comparison runs across medium/lower liquidity, sectors and mixed groups. Daily adjusted OHLCV observations and the ^GSPC benchmark come from the existing Yahoo Finance preparation route. Configured warm-up history starts before the modelling window; each observation is identified by Date and Ticker.

Universe	Size / composition
High Liquidity 30; Medium Liquidity 30; Lower Liquidity 30	30 securities in each named group
Sector Spread 30; Liquidity Barbell 30	30 securities; sector spread or high/low liquidity mix
Institutional Liquidity 60; Medium Small Liquidity 60; Medium Large Liquidity 60	60-security combinations
All Liquidity 90	Combined liquidity spectrum

Stock screening and feature construction

Stock-level screening is applied to training observations before the main feature panel is constructed. The sequence checks sample size, missingness, invalid observations and continuity relative to the market calendar.

Securities passing the selected screen become the feature-engineering universe.

The feature builder creates individual-stock measures, market-relative quantities and cross-sectional information.

The target-blind screen then examines missingness, invalid values, variance, duplicate construction and feature-to-feature correlation. This produces a common pool of features for subsequent target-specific analysis.

Targets and target-specific screens

Target columns are constructed from the price panel and placed alongside the features. The explicit research boundaries divide observations into training, validation and model-confirmation periods. Target-specific selection is performed on the training partition.

Target type	Screening sequence
All	Feature-target coverage and time stability
Continuous	Quantile spread and monotonicity, Pearson correlation and cross-sectional IC where applicable
Binary	Quantile spread, quantile monotonicity and Pearson correlation
Multiclass	Coverage and time-stability analysis

Each target starts with its own copy of the common feature pool. Selected_Features.txt records retained feature names by universe and target. Features_Targets_Data.db stores the research table with Date/Ticker, retained base columns, target columns and the union of selected predictors. Later stages load only the columns needed for the current target.

Optional intraday preparation

Intraday Conversion.py applies session-aware feature lookbacks and target horizons to intraday tables. It groups observations by ticker and session, identifies the required start-of-session history and reserves the forward rows needed by each target. Explicit override dictionaries describe custom column requirements. The initial daily research sequence proceeds directly to model fitting.

4 Point-in-time macro data and regime construction

Existing data source and availability-date alignment

Macro data is owned by research and comes from the existing macro creation pipeline. The database resolver uses macro_data.db, also recognising the legacy Macro_Data.db spelling. It discovers tables with Date and Macro PIT columns; unrelated tables are ignored. Multiple qualifying tables require an explicit table choice. A new independent macro source is not introduced.

The existing macro pipeline builds point-in-time features from ALFRED vintage information. Regimes use only columns beginning Macro PIT. The stock-data builder can attach those features using a backward availability-date join. It preserves stock row order and existing predictors, does not use a future macro row, and leaves unavailable early feature values missing.

Run macro creation before stock screening when macro predictors should enter newly fitted models. If stock screening ran first, rerun it after macro creation to include the new predictors. This ordering changes the information available to new research runs; it does not retroactively change already-frozen models.

Signals and economic interpretation

Family	Available signals used	States / minimum
Growth	Industrial production, retail sales and payrolls: 3M growth; real GDP: 1Q annualised growth; sentiment and housing starts: 3M versus prior 3M change	Expansion / Neutral / Contraction; 2 signals
Inflation	CPI, Core CPI and PCE: YoY acceleration over 3M; index-level rises alone do not classify inflation	Rising / Stable / Falling; 2 signals
Labour	Payroll 3M growth and YoY acceleration; inverse unemployment 1M change; unemployment 3M mean minus prior 12M minimum	Strong / Neutral / Deteriorating; 2 signals
Liquidity	M2 6M growth, 3M versus prior 3M change and YoY acceleration	Expanding / Neutral / Contracting; 1 signal
Consumer	Sentiment 1M/6M growth and 3M versus prior 3M change; retail sales 3M growth	Improving / Neutral / Weakening; 2 signals
Housing	Housing-starts 1M/6M growth, 3M versus prior 3M change and YoY acceleration	Improving / Neutral / Weakening; 2 signals

For each date, available signals vote +1, 0 or -1. The score is their mean; missing features do not invalidate the other votes. Most growth/change comparisons use a neutral band of 0.1 percentage points. Inflation, payroll and M2 acceleration and unemployment change use 0.05. The unemployment-gap vote is +1 below 0.20 percentage points, 0 from 0.20 to below 0.50, and -1 at or above 0.50.

```
score = mean(available votes)
score >= +0.34 -> positive family state
score <= -0.34 -> negative family state
otherwise -> neutral family state
fewer than required signals -> Unknown candidate
```

Causal confirmation and missing observations

The default confirmation period is five observations. The first usable state initializes immediately. A later candidate must persist for the required count before activation; the change is dated on the confirmation observation. Missing candidates retain the previously confirmed state and reset the pending candidate. Before any valid state exists, Unknown is retained.

Observation	Raw candidate	Active state with 3-observation confirmation
1	Expansion	Expansion
2	Contraction	Expansion
3	Contraction	Expansion
4	Contraction	Contraction

There is no retrospective relabelling of observations 2 and 3. Confirmation counts supplied observations: calendar-day input counts calendar days; trading-day input counts trading observations. Regime thresholds are fixed rules, not parameters fitted using future strategy performance.

Reusable outputs

Create Macro Regimes.py can generate regimes from the existing database without downloading data. The macro-data stage also saves Macro_Regimes.txt and Macro_Regimes_Daily.csv automatically. The required text artifact is a Python-literal dictionary with six family keys, all named states plus Unknown, and lists of inclusive ISO-date (start, end) tuples. Load it with ast.literal_eval, never eval.

```
{
'Growth': {'Expansion': [('2020-01-01', '2020-02-14')],
'Neutral': [], 'Contraction': [], 'Unknown': []},
...
}
Illustrative shape only; dates above are not estimated regimes.
```

The daily diagnostic table contains Date, the six active states, family scores and available-signal counts. The source database is not bundled with this report; no empirical regime chronology is claimed here.

5 Model fitting and chronological validation

Target-specific loading and estimator selection

The model-fitting stage reads the selected universe schema and processes one target at a time. The working frame contains Date, Ticker, the target and its selected features. Target/feature cleaning establishes the observations available for that task. Statistical target type determines the estimator families and evaluation metrics.

The grid catalogue spans baselines, linear and regularised models, robust regression, tree ensembles, boosting, support-vector methods, neural networks and nearest neighbours. Statistical target type determines the applicable regression or classification families. The original reference grid is retained below as documentation of the intended extensive profile; actual model lists remain editable in the code.

Setting group	Original reference values
Ridge alpha	0.01, 0.1, 1, 10, 100
Sparse-model alpha	0.00001, 0.0001, 0.001, 0.01
Classification C / L1 ratio	0.1, 1, 10 / 0.25, 0.75
Boosting learning rate / class weights	0.03, 0.1 / unweighted and balanced
Random forest	300 trees; depths 3, 6; leaf sizes 20, 50
XGBoost	300 estimators; depths 2, 4; regularised/subsampled settings
LightGBM	300 estimators; 7, 15 leaves; minimum child size 50
MLP / KNN	32-unit hidden layer / 20, 50, 100 neighbours

Configuration identity and validation history

A model configuration is identified by its estimator family and parameter dictionary. Each target stores fold-level observations and a search summary in its universe-specific validation database. The history supports the identification of previously evaluated configurations and the tracing of the metrics used in selection.

The principal table names are <target>__folds for individual fold results and <target>__search for the aggregated search leaderboard. Target identity, model parameters and fold information connect the stored observations.

Walk-forward calendar

The validation calendar is divided into consecutive windows of 63 observed dates. For each fold, the model is fitted on observations before the validation window. The training boundary is purged by the target's forward horizon. Any fitted preprocessing is learned from that training sample before being applied to the later validation observations.

The 2007-2012 validation period provides approximately 24 folds when the full daily calendar is represented.

The final window uses the remaining observed dates. Each target follows its own available validation calendar after its feature and target requirements are applied.

Progressive pruning

At each scheduled fold, a retained fraction and maximum count reduce the active configuration set. Surviving models continue through the remaining validation calendar; completing the pruning schedule does not end validation.

Completed fold	Retained fraction	Maximum configurations
3	95%	20,000
5	90%	5,000
7	80%	1,000
9	65%	300
11	50%	70
14	50%	15

Leaderboard and eligibility

Pruning determines which models remain active; it does not define the end of the validation calendar. A model that survives the fold-14 stage continues through the remaining folds. The research minimum of 15 validation folds is an eligibility condition used with the other performance and stability gates.

For each active model, the accumulated results describe both average prediction quality and variation between folds. Model selection therefore considers performance through time rather than a single successful interval.

Continuous tasks record RMSE, normalised RMSE, MAE, R^2 and Rank IC. Binary tasks record ROC AUC, PR AUC, log loss and F1-related measures. Multiclass tasks record Macro F1, balanced accuracy and log loss. The leaderboard combines the relevant fold statistics with model identity and portfolio-role information.

Stability / eligibility gate	Reference value
Continuous Rank IC standard deviation	Maximum 0.10
Binary ROC AUC standard deviation	Maximum 0.07
Multiclass Macro F1 standard deviation	Maximum 0.10

Stability / eligibility gate	Reference value
Completed validation folds	Minimum 15

The stored eligibility reason connects the decision to fold count, performance and variability. A minimum fold count is a qualification rule, not permission to stop the remaining scheduled folds.

6 Model confirmation and reusable metadata

Selection before confirmation

The model-confirmation stage reads eligible candidates from each target's validation results. Rank-one selection chooses the highest-ranked eligible configuration. An explicit selection mapping can instead identify a saved rank or an exact model/parameter combination. The selection is made before the model-confirmation block is scored.

The selected estimator is fitted using the preceding train and validation observations, with the target purge applied at the confirmation boundary. It then generates predictions for the 2013-2015 model-confirmation block.

Each target is evaluated using the metrics appropriate to its statistical type.

Statistical type	Stored test metrics	Original retention gates
Continuous	RMSE, MAE, R-squared, Rank IC	Rank IC at least 0.10
Binary	ROC AUC, PR AUC, log loss, F1	ROC AUC at least 0.60; PR AUC at least 0.20
Multiclass	Macro F1, balanced accuracy, log loss	Macro F1 at least 0.45

Final_Test_Results.db records target identity, selected estimator and serialised parameters, confirmation metrics and reusable downstream metadata. These measures describe target predictability; they are not portfolio-performance results.

Predictability Score

```
Continuous: P = max(Rank IC, 0)
Binary: P = 0.50 * [2 * max(ROC AUC - 0.50, 0)]
+ 0.30 * PR AUC + 0.20 * F1
Multiclass: P = Macro F1
```

Economic role and horizon

Broad role	Example fine-grained types
Ranking	ALPHA, RELATIVE_ALPHA, RISK_ADJUSTED_ALPHA, CROSS_SECTION_ALPHA
Direction	DIRECTION, DIRECTION_MULTICLASS, ALPHA_BINARY, BARRIER_ALPHA
Risk	VOLATILITY, DOWNSIDE_VOLATILITY, DOWNSIDE, TAIL_RISK, TAIL_EVENT
Opportunity	ABSOLUTE_MOVE, UPSIDE_VOLATILITY, UPSIDE_EVENT, RECOVERY, REVERSAL
Special	Excursion timing, EXECUTION, LIQUIDITY, MARKET_IMPACT, CORRELATION, REGIME

Canonical daily horizon keys include 1d, 5d, 20d, 60d, 120d and 252d where those targets exist. Role and horizon are descriptive metadata: downstream strategies can choose their own treatment without importing Horizon portfolio logic.

Quality Score

Absolute quality measures normalise the selected model's statistical evidence. Within each Portfolio Target Type, the relative Quality Score places the retained targets on a common scale. A single target, or a group with equal absolute quality, receives a neutral relative score of 0.5. The resulting quality information is carried into the cached signal construction.

```
For n retained targets in one target type:
lower = 0.5 / n; upper = 1 - lower
Q[j] = lower + (a[j] - min(a)) / (max(a) - min(a))
* (upper - lower)
One target, or equal absolute qualities: Q = 0.5
```

7 Confirmed model production and the output contract

The original combined project fitted selected models again inside Horizon research and cache creation, using different historical windows. The split moves all such fitting into research while preserving the exact training inputs and purge choices. The strategy loads checked fitted artifacts and has no fitting fallback.

Explicit request, production and resume

- A downstream preparation stage identifies the required selected target model and historical training partition. It checks for the corresponding artifact.
- If fits are absent, the stage writes all required model-only requests and stops before portfolio construction. Requests contain TRAIN rows only, never BACKTEST rows.
- Produce Confirmed Models.py processes those requests in the research project. The downstream stage then resumes and loads the confirmed fitted objects.
- Distinct historical phases or changed data/configurations may require another explicit research production pass. Future strategies can call `produce_models(training, selected, features, purge=...)` directly.

Artifact identity and provenance

The artifact key covers training values, column ordering/dtypes, selected feature order, model metadata and purge policy. A fitted artifact contains the estimator, scaler or encoder where relevant, target metadata and model summary. Its JSON manifest records training dates, row count, runtime information and a payload checksum. Missing, mismatched or corrupt artifacts are rejected. Pickle artifacts and request files are trusted local research outputs and should be consumed in the same dependency environment.

Output	Boundary role
Features_Targets_Data.db	Prepared feature/target data; target-specific loading
Selected_Features.txt	Universe- and target-keyed feature mapping
Validation_Model_Fits/<universe>.db	Fold histories and model-search summaries
Final_Test_Results.db	Confirmed model selection, parameters and target metrics
macro_data.db / legacy Macro_Data.db	Existing availability-safe macro observations/features

Output	Boundary role
Macro_Regimes.txt	Required frozen family/state interval definitions
Macro_Regimes_Daily.csv	Date-level states, scores and available-signal counts
confirmed_models/<identity>.pkl + .json	Fitted model objects and provenance/checksum manifest

8 Running the project and checking evidence

Stage key	Launcher
prepare	Prepare Research.py
data	Data_Creation_Screening.py
macro_data	Macro_Data_Creation_Screening.py
training	Model Fitting.py
model_confirmation	Best_Model_Test.py
macro_regimes	Create Macro Regimes.py
produce_models	Produce Confirmed Models.py

Run Research.py launches one selected stage. Prepare Research.py now creates the centrally configured research directory only. Intraday Conversion.py and Check Databases.py remain optional preparation/audit utilities. Each root script exposes editable settings; research does not stage strategy FX history or run account tests.

```
python -m pip install -e "Projects/Equity Model Research[test]"
python "Projects/Equity Model Research/Create Macro Regimes.py" \
--database /path/to/macro_data.db --table "Existing Table"
python "Projects/Equity Model Research/Produce Confirmed Models.py"
```

EQUITY_SELECTOR_DATA_DIR selects the research data directory; the default is this project's data/. EQUITY_SELECTOR_MODEL_DIR and EQUITY_SELECTOR_REQUEST_DIR optionally select model and request directories. Macro credentials are read from the ALFRED_API_KEY setting or FRED_API_KEY environment variable. Existing private data is not moved automatically.

Checks completed during the refactor

The shared validation run recorded 125 passing tests, with one pre-existing pandas fragmentation warning. Across both projects, 103 modules imported with SQLite/network access disabled. Static checks found no research-to-strategy imports and no strategy fitting calls. Deterministic tests verified artifact prediction equality, corruption rejection, sparse-signal regimes, literal-file determinism and backward macro alignment.

Every tested prefix of a macro series produced the same historical labels as the full series. These checks support the implementation's causal behavior; they are not a validation of an unavailable production macro dataset. Production databases and fitted historical artifacts were absent from the repository, so a full empirical research rerun was not performed.

Interpretation and future work

The project answers whether a selected prediction task has reproducible historical evidence under explicit research rules. It does not establish that a trading strategy is profitable. Large procedural stages and legacy package names remain for compatibility; further reorganisation should have its own production-data regression coverage. The explicit historical-fit request/resume boundary can later be used by strategies other than Horizon Score.

9 Reference universe definitions and source map

High Liquidity 30

AAPL, MSFT, AMZN, NVDA, META, TSLA, AMD, INTC, QCOM, MU, CSCO, ORCL, JPM, BAC, WFC, C, XOM, CVX, F, GM, T, VZ, PFE, JNJ, WMT, DIS, GE, HD, NFLX, GOOG

Medium Liquidity 30

ADI, MCHP, TXN, STX, WDC, PNC, USB, BK, STT, COF, CAT, DE, EMR, ETN, ITW, LOW, TGT, KR, BBY, DRI, AMGN, GILD, BIIB, BMY, CVS, OXY, EOG, SLB, HAL, VLO

Lower Liquidity 30

AIT, ARCB, BRC, CALM, CHCO, CPK, CNMD, FFIN, GATX, GBCI, GFF, HNI, HVT, JJSF, LANC, MGEE, MLAB, MMSI, MTRN, NRIM, NWN, RCKY, RELL, SCL, TNC, UVSP, WASH, WDFC, NATH, RES

Sector Spread 30

MSFT, ADI, RELL, JPM, PNC, CHCO, GE, CAT, AIT, AMZN, LOW, HVT, WMT, KR, JJSF, JNJ, AMGN, CNMD, XOM, OXY, RES, NEE, DUK, CPK, FCX, NUE, SCL, PLD, SPG, UHT

Liquidity Barbell 30

AAPL, MSFT, AMZN, NVDA, META, TSLA, JPM, BAC, XOM, CVX, WMT, JNJ, GE, HD, NFLX, AIT, ARCB, BRC, CALM, CHCO, CPK, CNMD, FFIN, GATX, HNI, JJSF, MGEE, MTRN, TNC, WDFC

Document provenance

Primary source: Equity Selector Project Report.pdf, dated 4 September 2026, supplied by the user. This report retains the model/data material from original sections 2, 4-8, 15-17 and appendices A, C and D, with an upstream-only boundary. Original sections 9-14 and portfolio formulas are documented in the companion Horizon Score Strategy Project Report.

Split-specific updates were checked against the locally refactored Equity Model Research README, equity_selector/artifacts.py, equity_selector/macro_regimes.py and the model launchers. The companion strategy report describes consumption of these outputs. Macro-regime construction and explicit fitted-model exchange are additions from the refactor, not empirical findings in the September 4 source report.